

Challenge Name: Operation: Ghost Researcher

1. Brief

The landing page frames the mission:

SUBJECT: Nguyễn Minh Khoa, self-described AI researcher.
An anonymous tip claims this researcher has hidden a sensitive credential somewhere in plain sight across his public online presence.

Four surfaces are named up front: **arXiv**, **X (Twitter)**, **GitHub**, **Pastebin**.

Two lines are the actual rules of engagement, and they matter more than the story:

"Start with what is most public: the paper on arXiv."
"Every site in this investigation is hosted on this same host — URLs found on one page work as relative links from any other."

That second line is the whole challenge design. Nothing here touches the real internet — `arxiv.org`, `github.com/minhkhhoa-ai`, `pastebin.com/ysv37wjrr` are all **simulated clones served from the same nginx instance**. Every external-looking URL you find must be re-mapped onto the challenge host. Searching the real arXiv or GitHub for these names returns nothing, and that dead end is the intended trap of the title: *someone who was never there*.

2. Recon

```
B="https://<instance>.222.255.138.122.nip.io"  
curl -sk "$B/" | head -100
```

The briefing links to `/arxiv/`. From there the challenge is a linked-list walk: each page contains exactly one handle or URL identifying the next hop.

The only technique needed is, at every page, dump the anchors before reading the prose, the mapped path is usually already sitting there:

```
curl -sk "$B/<page>" -o page.html  
grep -oE 'href="[^"]+"' page.html | sort -u  
sed -e 's/<[^>]*>/ /g' page.html | grep -v '^s*$' | head -80
```

3. The pivot chain

#	Simulated site	Path on host	Leak
1	Mission briefing	/	Start at arXiv, category cs.LG
2	arXiv cs.LG listing	/arxiv/	Paper arXiv:2605.99847 by Nguyễn Minh Khoa
3	Abstract page	/arxiv/abs/2605.99847	X handle @khoa_neuralnet
4	X profile	/x/khoa_neuralnet	GitHub minhkhoa-ai + "notes are in a paste" hint
5	GitHub profile	/gh/minhkhoa-ai	Pinned repo phantom-gradient-descent
6	Repo README	/gh/minhkhoa-ai/phantom-gradient-descent	Unlisted paste pastebin.com/ysv37wjrr
7	Pastebin	/p/ysv37wjrr	master_key_b64 → the flag

Step 2: arXiv listing

arXiv.org · Cornell University · All categories

arXiv.org > cs > cs.LG (Machine Learning)

Computer Science > Machine Learning (cs.LG)

Recent submissions · search the arXiv preprint repository.

Recent submissions in cs.LG (May 2026)

arXiv:2605.99821 [cs.LG]

A Note on Stochastic Gradient Methods for Federated Mean Estimation

M. Bachmann, J. Iyer (ETH Zürich)

We provide convergence guarantees for a class of federated SGD variants under heterogeneous client distributions...

arXiv:2605.99847 [cs.LG]

Efficient Neural Architecture Search via Phantom Gradient Descent

Nguyễn Minh Khoa, Trần Đức Anh, Sarah Mitchell

We propose Phantom Gradient Descent (PGD), a novel optimization technique for neural architecture search that reduces computational overhead by 73% compared to DARTS...

arXiv:2605.99863 [cs.LG]

Mixture-of-Experts Routing with Dynamic Capacity Constraints

L. Petrović, A. Sharma, T. Yamamoto (Stanford / KAIST)

Sparse MoE models suffer from load imbalance under non-uniform input distributions. We introduce a learned router with adaptive capacity factors...

arXiv:2605.99889 [cs.LG]

Constrained Diffusion Sampling on Discrete Manifolds

A. Khoury, E. Niedermayer (MILA)

We extend score-based diffusion models to settings where the target distribution is supported on a discrete combinatorial structure...

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/arxiv/ lists four cs.LG submissions for May 2026. Three are noise with only href="#" placeholders. Only one is clickable and only one carries the subject's name:

```
arXiv:2605.99847 [cs.LG]
Efficient Neural Architecture Search via Phantom Gradient Descent
Nguyễn Minh Khoa, Trần Đức Anh, Sarah Mitchell
```

A dead href="#" is the tell for a decoy throughout this challenge, the author only wired links on the intended path. Confirm it before clicking anything:

```
$ grep -oE 'href="[^\"]+"' arxiv.html | sort -u
href="#"
href="/arxiv/abs/2605.99847"
href="/arxiv/search"
```

Step 3: abstract page to X handle

arXiv.org Cornell University

arXiv.org > cs > cs.LG > arXiv:2605.99847

Efficient Neural Architecture Search via Phantom Gradient Descent

Nguyễn Minh Khoa¹, Trần Đức Anh¹, Sarah Mitchell²

¹Vietnam National Cyber Lab (fictional) · ²Independent Researcher
Submitted on 15 May 2026 (v1), last revised 18 May 2026 (this version, v2)
arXiv:2605.99847 [cs.LG] · License: CC BY 4.0

Abstract

We propose **Phantom Gradient Descent (PGD)**, a novel optimization technique for neural architecture search that reduces computational overhead by 73% compared to DARTS while maintaining competitive accuracy on CIFAR-10 and ImageNet benchmarks. Our approach leverages *ghost gradients* — synthetic gradient approximations computed from architectural perturbations — to efficiently explore the search space without full forward-backward passes. Experiments demonstrate state-of-the-art results on NAS-Bench-201 with only 0.3 GPU-days of search time. Code and configurations are available on the corresponding author's GitHub.

Author Information & Links

Nguyễn Minh Khoa (corresponding author) — PhD candidate, Vietnam National Cyber Lab.
Research interests: Neural Architecture Search, AutoML, efficient deep learning systems.

🐦 X (Twitter): @khoa_neuralnet
✉ Contact: khoa.nguyen [at] vncyber-lab.example

Code, pre-trained weights and search logs are available via the author's GitHub, linked from his X profile.

arXiv preprint server · Cornell University · Privacy Policy · Contact

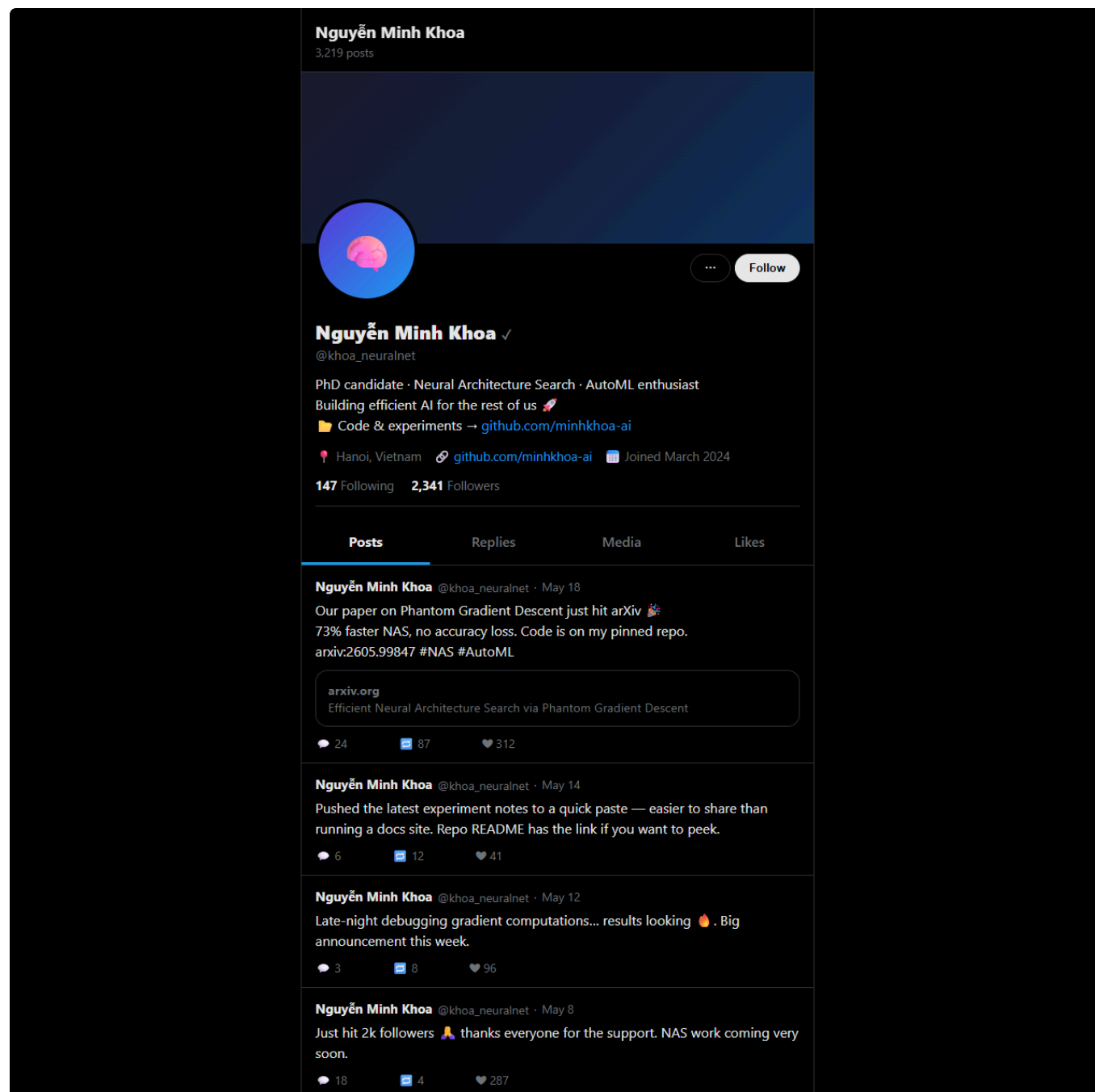
/arxiv/abs/2605.99847 carries an "Author Information & Links" sidebar:

Nguyễn Minh Khoa (corresponding author) — PhD candidate, Vietnam National Cyber Lab.

 X (Twitter): @khoa_neuralnet
 Contact: khoa.nguyen [at] vncyber-lab.example
Code, pre-trained weights and search logs are available via the author's GitHub,
linked from his X profile.

That last sentence is an explicit routing instruction: GitHub is *not* named here, you have to go through X to get it.

Step 4: X profile to GitHub



Nguyễn Minh Khoa
3,219 posts



Nguyễn Minh Khoa ✓
@khoa_neuralnet
PhD candidate · Neural Architecture Search · AutoML enthusiast
Building efficient AI for the rest of us 🚀
📁 Code & experiments → github.com/minhkhoa-ai
📍 Hanoi, Vietnam 🔗 github.com/minhkhoa-ai 📅 Joined March 2024
147 Following 2,341 Followers

Posts Replies Media Likes

Nguyễn Minh Khoa @khoa_neuralnet · May 18
Our paper on Phantom Gradient Descent just hit arXiv 🎉
73% faster NAS, no accuracy loss. Code is on my pinned repo.
[arxiv:2605.99847](https://arxiv.org/abs/2605.99847) #NAS #AutoML

[arxiv.org](https://arxiv.org/abs/2605.99847)
Efficient Neural Architecture Search via Phantom Gradient Descent

24 87 312

Nguyễn Minh Khoa @khoa_neuralnet · May 14
Pushed the latest experiment notes to a quick paste — easier to share than running a docs site. Repo README has the link if you want to peek.

6 12 41

Nguyễn Minh Khoa @khoa_neuralnet · May 12
Late-night debugging gradient computations... results looking 🔥. Big announcement this week.

3 8 96

Nguyễn Minh Khoa @khoa_neuralnet · May 8
Just hit 2k followers 🙌 thanks everyone for the support. NAS work coming very soon.

18 4 287

/x/khoa_neuralnet — note /twitter/... 404s; the anchor in the abstract page gives the right path. Bio and pinned links:

📁 Code & experiments → github.com/minhkhoa-ai
🔗 github.com/minhkhoa-ai
📍 Hanoi, Vietnam · Joined March 2024 · 2,341 Followers

The timeline post from **May 14** the same date as the paste, hints two hops ahead:

"Pushed the latest experiment notes to a quick paste, asier to share than running a docs site. Repo README has the link if you want to peek."

Path-mapping gotcha. The displayed URL is `github.com/minhkhoa-ai`, but naively translating it to `/github/minhkhoa-ai` returns 404. The real mapping is `/gh/`:

```
$ for p in /gh/minhkhoa-ai /github/minhkhoa-ai /github.com/minhkhoa-ai
/minhkhoa-ai; do
    printf "%-26s " "$p"; curl -sk -o /dev/null -w "%{http_code}\n" "$B$p"
done
/gh/minhkhoa-ai           200
/github/minhkhoa-ai       404
/github.com/minhkhoa-ai   404
/minhkhoa-ai              404
```

The 404 page even warns *"most paths are case-sensitive"*. Rather than guessing prefixes, read the anchor the page already gives you, `href="/gh/minhkhoa-ai"`. Same pattern later: `pastebin.com/ysv37wjr` → `/p/ysv37wjr`.

Step 5: GitHub profile to repo

minhkhoea-ai



Nguyễn Minh Khoa

minhkhoea-ai

PhD candidate · Neural Architecture Search · AutoML enthusiast.

 Vietnam National Cyber Lab (fictional)

 Hanoi, Vietnam

 @khoea_neuralnet

 khoa.nguyen [at] vncyber-lab.example

Pinned

phantom-gradient-descent

Public

Official PyTorch implementation of *Phantom Gradient Descent* (arXiv:2605.99847).

• Python ⭐ 342 📄 47

nas-bench-utils

Public

Light-weight utility scripts for parsing and visualising NAS-Bench-201 / -301 results.

• Python ⭐ 18 📄 4

cifar-baselines

Public

Personal collection of CIFAR-10/100 training baselines (ResNet, DenseNet, EfficientNet).

• Python ⭐ 6 📄 1

/gh/minhkhoea-ai shows three repos. Two are decoys with href="#" (nas-bench-utils, cifar-baselines); the pinned phantom-gradient-descent is the only live link, and it matches the paper title.

Step 6: README to paste

minhkhoe-ai / phantom-gradient-descent Public

Watch 28

Fork 47

Star 342

About

Official PyTorch implementation of *Phantom Gradient Descent* — efficient neural architecture search via ghost gradients.

neural-architecture-search

automl

pytorch

deep-learning

main · last commit 2 days ago by minhkhoe-ai

configs	add NAS-Bench-201 config	3 days ago
models	refactor PGD core loop	2 days ago
utils	add gradient logging helpers	5 days ago
.gitignore	initial commit	2 weeks ago
LICENSE	add MIT license	2 weeks ago
README.md	link experiment notes	2 days ago
requirements.txt	pin torch 2.3	1 week ago
search.py	add PGD search loop	2 days ago
train.py	add training pipeline	4 days ago

README.md

Phantom Gradient Descent

Efficient Neural Architecture Search via Ghost Gradients

This repository contains the official PyTorch implementation of our paper "**Efficient Neural Architecture Search via Phantom Gradient Descent**" (arXiv:2605.99847, 2026).

Key Results

- **73% faster** than DARTS on NAS-Bench-201
- **97.42% accuracy** on CIFAR-10 (comparable to SOTA)
- Only **0.3 GPU-days** of search time

Installation

```
git clone https://github.com/minhkhoe-ai/phantom-gradient-descent.git
cd phantom-gradient-descent
pip install -r requirements.txt
```

Usage

```
# Run architecture search
python search.py --dataset cifar10 --method pgd --epochs 50

# Train discovered architecture
python train.py --arch discovered_arch.json --epochs 600
```

Experiment Notes

Full experiment logs, hyper-parameter sweeps and one-off configurations are kept in a shared workspace paste rather than cluttering this repo:

🔗 Internal research notes (Pastebin, unlisted):
<https://pastebin.com/ysv37wjr>

Citation

@article{nguyen2026phantom,

```
title = {Efficient Neural Architecture Search via Phantom Gradient Descent},
author = {Nguyen, Minh Khoa and Tran, Duc Anh and Mitchell, Sarah},
journal= {arXiv preprint arXiv:2605.99847},
year   = {2026}
}
```

License

MIT — see [LICENSE](#) for details.

/gh/minhkhoea-ai/phantom-gradient-descent renders a full README. Under

Experiment Notes:

Full experiment logs, hyper-parameter sweeps and one-off configurations are kept in a shared workspace paste rather than cluttering this repo:

 Internal research notes (Pastebin, unlisted): <https://pastebin.com/ysv37wjr>

Mapping pastebin.com/<id> → </p/<id>> gives </p/ysv37wjr>.

Step 7: the paste

PGD Research Notes — Internal

Posted by minhkhoe-ai Created: May 14, 2026 Visibility: Unlisted Syntax: Plain text

```
1 # 
2 # Phantom Gradient Descent - Internal Research Notes
3 # Author: Nguyễn Minh Khoa
4 # Date : 2026-05-14
5 # Status: CONFIDENTIAL - internal use only
6 # 
7 
8 ## Recent experiment runs
9 Run #47 | CIFAR-10 | 50 epochs | lr=0.025 | acc=97.42% ✓
10 Run #48 | ImageNet (10%) | 100 epochs | lr=0.012 | top1=76.8% ✓
11 Run #49 | NAS-Bench-201 | search | 0.3 GPU-d | best ✓
12 
13 ## API tokens (rotate before public release!)
14 wandb_api_key = "wk-3f8a9b2c1d4e5f6a7b8c9d0e" # dummy
15 hf_token = "hf_QwErTyUiOpAsDfGhJkLzXcVbNm" # dummy
16 
17 # Project master credential - base64-encoded so it doesn't trip secret scanners
18 master_key_b64 = "ZmxhZ3syOTQ4M2NkNy04MjI2LTQ2YTMTOWQ0MC1kMmM3NGU5ZDA3ZTZ9"
19 
20 ## TODO before camera-ready
21 [ ] Rotate every credential listed above
22 [ ] Strip experiment logs of internal hostnames
23 [ ] Move this paste to a private workspace
```

Raw paste · Unlisted · Created May 14, 2026 · Pastebin.com (simulated for OSINT challenge)

```
## API tokens (rotate before public release!)
wandb_api_key = "wk-3f8a9b2c1d4e5f6a7b8c9d0e" # dummy
hf_token = "hf_QwErTyUiOpAsDfGhJkLzXcVbNm" # dummy

# Project master credential - base64-encoded so it doesn't trip secret
scanners
master_key_b64 = "ZmxhZ3syOTQ4M2NkNy04MjI2LTQ2YTMTOWQ0MC1kMmM3NGU5ZDA3ZTZ9"
```

The `wandb` and `hf` tokens are explicitly annotated `# dummy`, decoys shaped like real credentials, there to bait a premature submission. The comment above `master_key_b64` is a gift: it names both the encoding and the motive.

4. Flag

```
echo "ZmxhZ3syOTQ4M2NkNy04MjI2LTQ2YTMtOWQ0MC1kMmM3NGU5ZDA3ZTZ9" | base64 -d
```